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Professional Summary

AI Engineer specializing in production-ready Generative AI systems, with a strong focus on agentic architectures, Retrieval-Augmented Generation (RAG), LLM orchestration, semantic retrieval, and document intelligence. Experienced in designing cloud-native AI platforms using Azure OpenAI, Azure AI Search, Semantic Kernel agents, custom tools, and containerized microservices, while also bringing a solid background in machine learning, data science, and scalable cloud deployments across healthcare, insurance, manufacturing, marketing, finance, education, and scientific research domains.

Core Technologies

Generative AI: RAG, Agentic AI, Semantic Kernel, LangChain, LangGraph, DSPy, Azure OpenAI, Amazon Bedrock, Langfuse, DeepEval, Prometheus

Cloud & Data: Azure, Azure AI Search, Azure Databricks, AWS, SageMaker, Lambda, Step Functions, S3, SQS, EC2, Athena, DynamoDB

Engineering: Python, Node.js, Docker, SQL, PostgreSQL, MongoDB, REST APIs, Microservices, C#, Julia

ML: TensorFlow, Keras, Scikit-Learn, document classification, entity recognition, OCR, embeddings, vector search

Experience

Senior AI Engineer at Publicis Sapient

Sep 2024 – Present

Healthcare, Artificial Intelligence, Manufacturing, Big Data, Marketing

- Designed and developed enterprise-grade **Retrieval-Augmented Generation (RAG)** systems for document intelligence and knowledge discovery across complex client environments.
- Built agentic AI solutions using **Microsoft Semantic Kernel agents**, custom tools, and orchestrated multi-agent workflows, enabling assistants to reason, retrieve information, call tools, and execute domain-specific tasks.
- Implemented a cloud-native **microservice architecture** where individual agents and groups of agents are deployed as isolated containerized services, improving scalability, reliability, and maintainability.
- Developed retrieval workflows using **Azure AI Search** for vector, hybrid, semantic, and metadata-aware search, enabling grounded and contextually relevant responses from enterprise knowledge bases.
- Created custom ingestion pipelines using **Mistral OCR**, document parsing, chunking, metadata enrichment, embedding generation, and automated indexing into Azure AI Search.
- Integrated **Azure OpenAI** and other LLM providers into modular RAG and agentic workflows, allowing flexible model usage and client-specific deployment configurations.
- Built custom tools for AI agents, including retrieval tools, database access tools, API integrations, web search, and business-logic execution components.
- Implemented document versioning, re-indexing, and synchronization logic to keep source documents, processed chunks, embeddings, and search indexes consistent across updates.
- Optimized ingestion and retrieval performance through parallel processing, chunking strategy improvements, metadata modeling, and search relevance tuning.
- Developed AI assistants for enterprise and sales teams, combining RAG, agentic reasoning, structured database access, and external search capabilities to support decision-making and operational efficiency.
- Technologies: Python, Azure, Azure AI Search, Azure OpenAI, Semantic Kernel, Mistral OCR, Docker, PostgreSQL, MongoDB, AWS, Azure Databricks

Machine Learning Consultant at Scrubs

Jan 2025 – Aug 2025

Healthcare, Artificial Intelligence, Insurance

- Developed and integrated LLM-based features, including RAG systems and agentic workflows using LangChain and LangGraph, to automate insurance processes such as OCR, policy rule extraction, claim assessment, and document categorization.
- Collaborated on backend development of a scalable Node.js application for real-time ML services, integrating AWS services such as S3, Lambda, EC2, and SQS into the platform architecture.
- Created evaluation frameworks for LLM components using Langfuse, DeepEval, and Prometheus, improving

observability, traceability, and response quality in insurance operations.

- Technologies: Python, Node.js, MongoDB, AWS, LangChain, LangGraph, Amazon Bedrock

Machine Learning Consultant at Falcony AI

Jan 2025 – May 2025

Artificial Intelligence, Science

- Developed LLM-powered features for scientific writing, enabling web search, literature retrieval from vector databases, and multi-step agentic workflows using Python and DSPy.
- Enhanced RAG pipelines by integrating external search APIs such as Serper and Tavily to improve relevance and coverage of scientific literature in generated outputs.
- Contributed to AWS-based deployment by configuring EC2, Lambda, SQS, and DynamoDB components for scalable and resilient research assistant tools.
- Technologies: Python, AWS, DynamoDB, DSPy, Vector Databases, Serper, Tavily

Machine Learning Engineer at EPAM Systems

May 2022 – Sep 2024

Artificial Intelligence, Big Data, Marketing, Market Research

- Pioneered the Machine Learning Engineer role on client projects, implementing end-to-end SageMaker pipelines for model training, validation, versioning, and inference.
- Leveraged AWS services including SageMaker, Lambda, Step Functions, S3, DynamoDB, and Athena to deploy models, orchestrate ML workflows, and generate client-facing reports.
- Supported production ML systems across marketing and market research use cases, combining model development, cloud infrastructure, and analytics workflows.
- Technologies: Python, AWS, SageMaker, TensorFlow, Keras, Scikit-Learn, Athena, DynamoDB

Teaching Assistant at Faculty of Technical Sciences

Oct 2019 – Nov 2023

University of Novi Sad, Education, Artificial Intelligence

- Educated students on system modeling, simulation, distributed control systems, SCADA systems, and machine learning.
- Conducted research in Artificial Intelligence while pursuing a Ph.D. in Computer Science and Automation, specializing in Machine Learning and Intelligent Systems.
- Technologies: Python, SQL, MongoDB, C#, TensorFlow, Keras, Julia

Data Scientist at Mistho

Oct 2022 – Sep 2023

Artificial Intelligence, Personal Finance, Financial Services

- Pioneered the Data Science function as the first data science hire, establishing core ML capabilities for income verification workflows.
- Led Document Intelligence development within the Manual Upload feature, enabling efficient extraction and verification of data from payslips.
- Implemented and deployed production models for entity recognition and document classification.
- Technologies: Python, Node.js, MongoDB, AWS, TensorFlow, Keras

Freelance Software Developer and Machine Learning Engineer at Upwork

Feb 2019 – Oct 2022

Artificial Intelligence, Machine Learning, Data Science, Big Data

- Collaborated with international clients to transform software ideas into production-ready products across web, backend, and ML use cases.
- Delivered projects as both a software developer and machine learning engineer, contributing to backend services, computer vision models, and automation systems.
- Technologies: Python, SQL, MongoDB, C#, TensorFlow, Scikit-Learn, Keras

Selected Projects and Research

Determining Gaze Behaviour Patterns in On-Screen Testing

Publication

- Conducted machine learning and eye-tracking research in collaboration with professors Savić G. and Segeđinac M., published in the *Journal of Educational Computing Research*.

Multipurpose Remote Monitoring System Based on Microservice Architecture

Publication

- Collaborated on research published at the International Conference on Electrical, Electronic and Computing Engineering, focusing on a microservice-based remote monitoring system.

Automation of Framing Football Matches

- Worked as a software developer and machine learning engineer on a computer vision system for automatic

football match framing, including ball detection using deep neural networks and automated YouTube stream camera selection.

Automation of Recording Sport Events

- Developed a web application for tennis match scheduling and an embedded system running on a mini PC to automatically control cameras, record video, and upload footage.

Automatic Segmentation of ECG Signals Using Deep Learning

- Worked as a scientific researcher on a deep learning project using U-Net neural networks to segment ECG signals into clinically relevant regions.

Analysis of Eye-Tracking Data Using Machine Learning Techniques

- Delivered an end-to-end research workflow, from eye-tracking data collection to advanced preprocessing, visualization, and clustering analysis.

Education

Ph.D. in Machine Learning and Intelligent Systems <i>Faculty of Technical Sciences, University of Novi Sad</i>	Oct 2020 – Present
Master's Degree in Machine Learning and Intelligent Systems <i>Faculty of Technical Sciences, University of Novi Sad</i>	Oct 2019 – Aug 2020
Undergraduate Degree in Software Engineering <i>Faculty of Technical Sciences, University of Novi Sad</i>	Oct 2015 – Aug 2019

Certifications

MCP: Build Rich-Context AI Apps with Anthropic <i>DeepLearning.AI</i> Skills: Large Language Models (LLM), Model Context Protocol (MCP)	Jul 2025
Secure Coding - Ensuring Safe Deployment of Code <i>Udemy</i> Skills: Secure Coding, Software Deployment	Jul 2025
Oracle Cloud Infrastructure 2024 Generative AI Certified Professional <i>Oracle</i> Skills: Generative AI, Large Language Models (LLM), Oracle Cloud Infrastructure	Nov 2024
Oracle Cloud Infrastructure 2024 Certified Data Science Professional <i>Oracle</i> Skills: Data Science, Machine Learning, Oracle Cloud Infrastructure	Oct 2024
Oracle Cloud Infrastructure 2024 Certified Foundations Associate <i>Oracle</i> Skills: Oracle Cloud Infrastructure, Cloud Computing	Oct 2024
Oracle Machine Learning using Autonomous Database 2024 Certified Associate <i>Oracle</i> Skills: Machine Learning, Python, Oracle Autonomous Database	Oct 2024
LangGraph - Develop LLM Powered AI Agents with LangGraph <i>Udemy</i> Skills: Large Language Models (LLM), Agentic AI	Oct 2024
LangChain - Develop LLM Powered Applications with LangChain <i>Udemy</i> Skills: Large Language Models (LLM), RAG Applications	Oct 2024
Oracle Cloud Infrastructure 2024 Certified AI Foundations Associate <i>Oracle</i> Skills: Artificial Intelligence, Oracle Cloud Infrastructure	Sep 2024
NVIDIA DLI Certificate - Fundamentals of Deep Learning for Computer Vision <i>NVIDIA</i> Credential ID: 1fa7fc174522490ba8472a57d1d135ef Skills: Deep Learning, Computer Vision	Dec 2020